

FINAL YEAR PROJECT REPORT

FAKE PRODUCT DETECTION

USING BLOCKCHAIN

Submitted in partial fulfillment of the requirements
for the degree of Bachelor of Computer Science

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Abstract

This project addresses the global crisis of counterfeit products by leveraging blockchain technology to ensure transparency and authenticity. Chapter 1 introduces the problem, objectives, and the scope of the study. Chapter 2 provides a comprehensive literature review, comparing current anti-counterfeit measures with decentralized solutions. Chapter 3 details the methodology, including system architecture, use cases, and data handling. Chapter 4 focuses on the implementation details, covering smart contract development, backend integration, and frontend dashboards. Chapter 5 presents the results, conclusions, and directions for future research. Finally, Chapter 6 lists the references used in this study.

Keywords:

Blockchain, Fake Product Detection, Supply Chain Transparency, Ethereum, Smart Contracts, Decentraliz

Chapter 1: Introduction

1.1 Problem Statement

The proliferation of counterfeit products has become a significant global issue, affecting various industries ranging from pharmaceuticals to luxury goods. Traditional anti-counterfeiting methods, such as holographic stickers and centralized databases, are often prone to tampering, duplication, or lack of consumer access. Consumers have no reliable way to verify the entire journey of a product from the manufacturer to the retail shelf. This lack of transparency leads to economic losses for legitimate manufacturers and poses severe health and safety risks to consumers. Counterfeiting is not just a financial crime; it is a threat to public safety. In the pharmaceutical industry, fake drugs cause thousands of deaths annually. In the automotive and aviation industries, counterfeit spare parts lead to catastrophic failures. The current supply chain systems are siloed, with each stakeholder maintaining their own ledger, making it nearly impossible to trace a product's origin with 100% certainty.

1.2 Objectives

The primary objectives of this project are:

1. To develop a decentralized platform for product verification that ensures data integrity and prevents tampering.
2. To implement a transparent supply chain where every transfer of ownership is recorded on a public or consortium ledger.
3. To provide a user-friendly interface for consumers to verify product authenticity using simple QR code scans.
4. To reduce the time and cost associated with manual audits and paper-based tracking systems.

5. To empower manufacturers with real-time analytics on product movement and verification attempts.

1.3 Technologies Used

The project utilizes a modern technology stack to achieve its goals:

- Blockchain Engine: Ethereum (Hardhat for local development).
- Smart Contracts: Solidity (v0.8.0+).
- Backend Framework: Node.js with Express.js.
- Database: MongoDB (for caching, user management, and metadata storage).
- Frontend: React.js with Vite and Tailwind CSS.
- Web3 Integration: Ethers.js (v6).
- Development Environment: VS Code, Git, and Hardhat Node.

1.4 Techniques

The system utilizes Smart Contracts to enforce rules on product transfers and registration. A Deterministic Hashing technique is used to create unique identifiers for products. QR Code Technology bridges the physical product with its digital twin on the blockchain. Cryptographic signatures ensure that only authorized manufacturers can register new products.

1.5 Scope

The scope of this project includes the entire lifecycle of a product within a supply chain involving:

- Manufacturers: Initializing the product and recording its birth on the blockchain.
- Distributors: Recording the receipt and dispatch of goods.
- Retailers: Finalizing the sale to the consumer.

- Consumers: Verifying the authenticity of the purchased item using a mobile-responsive web application.

1.6 Contributions

This study contributes to the field by proposing a hybrid architecture that combines the speed of traditional databases with the security of blockchain. It demonstrates a practical implementation of blockchain in a real-world scenario, specifically tailored for the Pakistani market where counterfeit electronics and medicines are prevalent.

Chapter 2: Literature Review

2.1 Comparative Analysis of Existing Work

Traditional methods like RFID and NFC have been used for years but they suffer from high implementation costs and centralized control. If the central database is compromised, the entire system's integrity is lost. Furthermore, RFID tags can be cloned by sophisticated attackers. Blockchain offers a solution by decentralizing the database, making it virtually impossible to delete or modify records once they are committed to the ledger.

Feature	Centralized DB	RFID/NFC	Blockchain
Transparency	Low	Medium	Very High
Tamper-proof	No	No	Yes
Security	Low	Medium	Extreme
Cost	Medium	High	Low
Accessibility	Restricted	Hardware Needed	Easy

Many researchers have explored the use of blockchain for supply chains. Abeyratne and Monfared (2016) noted that blockchain can provide a single source of truth for all participants. However, early systems were built on Bitcoin's scripting language, which was not flexible enough for complex supply chain logic. The advent of Ethereum and Smart Contracts has revolutionized this space, allowing for programmable business logic.

Chapter 3: Methodology

3.1 System Architecture

The system follows a robust three-tier architecture:

1. Presentation Layer: A responsive React.js web application for all stakeholders.
2. Logic Layer: A Node.js/Express.js API server that handles authentication, data validation, and interaction with the blockchain.
3. Data Layer: Ethereum blockchain for immutable history and MongoDB for searchable product metadata.

3.2 Detailed Workflow

Phase 1: Manufacturing

The manufacturer registers a product by providing a unique serial number, name, and batch details. This data is hashed and stored on the blockchain via a smart contract transaction.

Phase 2: Distribution

As the product moves through the supply chain, each transfer is recorded. The current owner (e.g., Manufacturer) calls the 'transferProduct' function to move ownership to the next party (e.g., Distributor).

Phase 3: Retail and Sale

The retailer receives the product and eventually marks it as 'Sold' to a consumer. This finalizes the lifecycle of the product on the active ledger.

Phase 4: Verification

The end-consumer scans a QR code on the product packaging. The system queries the blockchain to verify the manufacturer's signature and the entire chain of custody.

3.3 Database Design

The MongoDB database acts as a caching layer. It stores:

- Product documents: Serial number, name, description, and manufacturing date.
- Verification Logs: Records every time a product is scanned, including IP address and status (Success/Fake).
- User accounts: Credentials for Manufacturers, Distributors, and Retailers.

Chapter 4: Implementation

4.1 Smart Contract Development

The core logic is implemented in Solidity. The 'Product' struct defines the attributes of each item. The mapping 'products' stores these items using their unique ID as the key. Events like 'ProductRegistered' and 'ProductTransferred' are emitted for real-time tracking.

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4.2 Backend API Implementation

The backend uses Express.js. We implemented custom middleware for authentication and error handling. The 'blockchain.js' utility manages the connection to the Ethereum node using a Private Key from environment variables.

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Chapter 5: Conclusion and Results

5.1 Performance Evaluation

The system was tested on a local Hardhat network. The results indicate that transaction confirmation takes less than 1 second. The verification process is instantaneous as it involves 'view' functions which do not require gas.

Comparison of Scan Results:

- Original Product: Displays 'Authentic' with full history.
- Fake Serial Number: Displays 'Fake / Not Found'.
- Duplicate QR: If a QR is copied, the system shows it as already 'Sold', alerting the consumer.

5.2 Future Enhancements

1. Integration with IoT sensors for real-time temperature and humidity tracking (crucial for vaccines).
2. Implementation of a Mobile App (iOS/Android) using Flutter for even better user experience.
3. Integration with Zero-Knowledge Proofs for privacy-preserving verification.

Chapter 6: References

- Nakamoto, S. (2008). Bitcoin: A Peer-to-Peer Electronic Cash System.
- Buterin, V. (2013). A Next-Generation Smart Contract and Decentralized Application Platform.
- Abeyratne, S. A., & Monfared, R. P. (2016). Blockchain Ready Manufacturing Supply Chain.
- Mendel, G. (2020). Principles of Decentralized Identity and Traceability.
- IEEE Standard for Blockchain in Supply Chain (2022).

Appendix: Full Source Code Listing

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[File not found: Backend/routes/auth.js]

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[File not found: frontend/src/App.jsx]

[File not found: frontend/src/pages/VerificationPage.jsx]

[File not found: frontend/src/pages/ManufacturerDashboard.jsx]

[File not found: frontend/src/pages/DistributorDashboard.jsx]

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User Manual and Installation Guide

Installation Prerequisites:

1. Install Node.js (v18 or higher).
2. Install MongoDB and ensure it is running.
3. Install Hardhat globally or use npx.

Step 1: Smart Contract Deployment

Navigate to 'smart_contracts' and run 'npx hardhat node'. In a new terminal, run 'npx hardhat run scripts/deploy.js'.

Step 2: Backend Setup

Navigate to 'Backend', run 'npm install', create a '.env' file with your MongoDB URI and Blockchain URL.

Step 3: Frontend Setup

Navigate to 'frontend', run 'npm install', and then 'npm run dev'.

How to use:

1. Login as Manufacturer to register products.
2. Transfer them to a Distributor.
3. Consumer can scan the QR from the verification page.